

8_puzzle problem:

1.DFS

```
Step 0:  
1 2 3  
... 8 4 5  
7 0 6
```

```
Step 1:  
1 2 3  
8 0 5  
7 4 6
```

```
Step 2:  
1 0 3  
8 2 5  
7 4 6
```

```
Step 3:  
0 1 3  
8 2 5  
7 4 6
```

```
Step 4:  
8 1 3  
0 2 5  
7 4 6
```

```
Step 5:  
8 1 3  
7 2 5  
0 4 6
```

```
Step 6:  
8 1 3  
7 2 5  
4 0 6
```

```
Step 7:  
8 1 3  
7 0 5  
4 2 6
```

```
Step 8:  
8 0 3  
... 7 1 5  
4 2 6
```

```
Step 9:  
8 3 0  
7 1 5  
4 2 6
```

```
Step 10:  
8 3 5  
7 1 0  
4 2 6
```

```
Step 11:  
8 3 5  
7 0 1  
4 2 6
```

```
Step 12:  
8 3 5  
7 2 1  
4 0 6
```

```
Step 13:  
8 3 5  
7 2 1  
0 4 6
```

```
Step 14:  
8 3 5  
0 2 1  
7 4 6
```

```
Step 15:  
8 3 5  
2 0 1  
7 4 6
```

```
Step 16:  
8 3 5  
2 1 0  
7 4 6
```

```
Step 17:  
8 3 0  
2 1 5  
7 4 6
```

```
Step 18:  
8 0 3  
2 1 5  
7 4 6
```

```
Step 19:  
8 1 3  
2 0 5  
7 4 6
```

```
Step 20:  
8 1 3  
2 4 5  
7 0 6
```

```
... Step 21:  
8 1 3  
2 4 5  
7 6 0
```

```
Step 22:  
8 1 3  
2 4 0  
7 6 5
```

```
Step 23:  
8 1 3  
2 0 4  
7 6 5
```

```
Step 24:  
8 1 3  
0 2 4  
7 6 5
```

```
Step 25:  
0 1 3  
8 2 4  
7 6 5
```

```
Step 26:  
1 0 3  
8 2 4  
7 6 5
```

```
Step 27:  
1 2 3  
8 0 4  
7 6 5
```

```
===== IDS Solution =====
```

```
IDS found solution at depth limit: 3
```

```
IDS Path length: 3 moves
```

```
Step 0:  
1 2 3  
8 4 5  
7 0 6
```

```
Step 1:  
1 2 3  
8 4 5  
7 6 0
```

```
Step 2:  
1 2 3  
8 4 0  
7 6 5
```

```
Step 3:  
1 2 3  
8 0 4  
7 6 5
```